

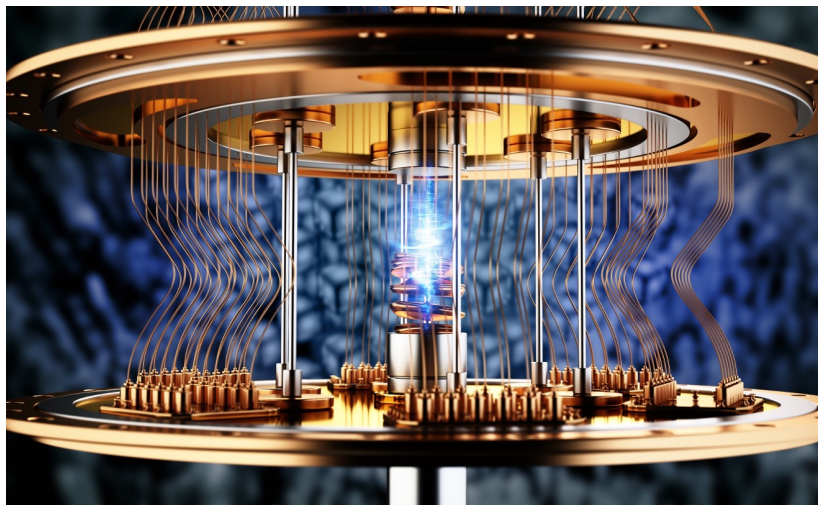
Science for AI 科学赋能人工智能

June 5

Tailin Wu, Westlake University



AI + Quantum computing and Aerospace



Quantum computing



Aerospace





Classical computation:

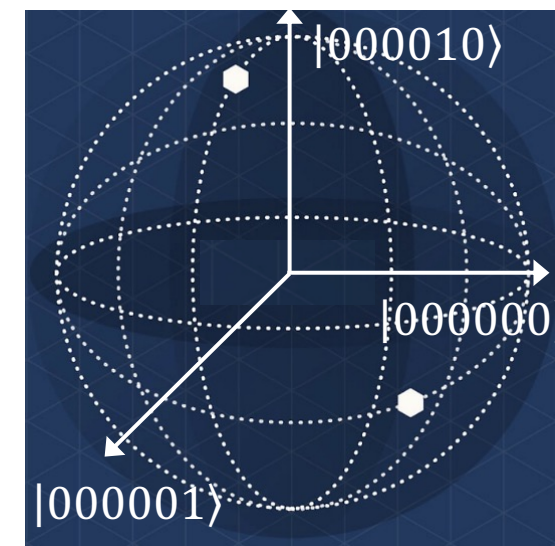
- *Classical-computation*(classical state)
- E.g., $f(101110) = 000001$

Quantum computation:

- *Quantum-computation*(quantum state)
- E.g., $U \left(\frac{1}{\sqrt{3}} |000000\rangle + \frac{1}{\sqrt{3}} |001001\rangle + \frac{1}{\sqrt{3}} |101011\rangle \right) = \sqrt{0.02} |010000\rangle + \sqrt{0.98} |111111\rangle$

Here U is a unitary matrix, i.e. “rotation” in the 2^n dimensional space

The power of quantum computation is that it can perform **simultaneous operation** on 2^n classical states



a quantum state



A good policy $\pi_1(a|s)$:

	1	2	3	4	5
1					
2					
3					
4					
5					

Corresponding state value: $\pi_1(a|s)$:

	1	2	3	4	5
1	3.5	3.9	4.3	4.8	5.3
2	3.1	3.5	4.8	5.3	5.9
3	2.8	2.5	10.0	5.9	6.6
4	2.5	10.0	10.0	10.0	7.3
5	2.3	9.0	10.0	9.0	8.1

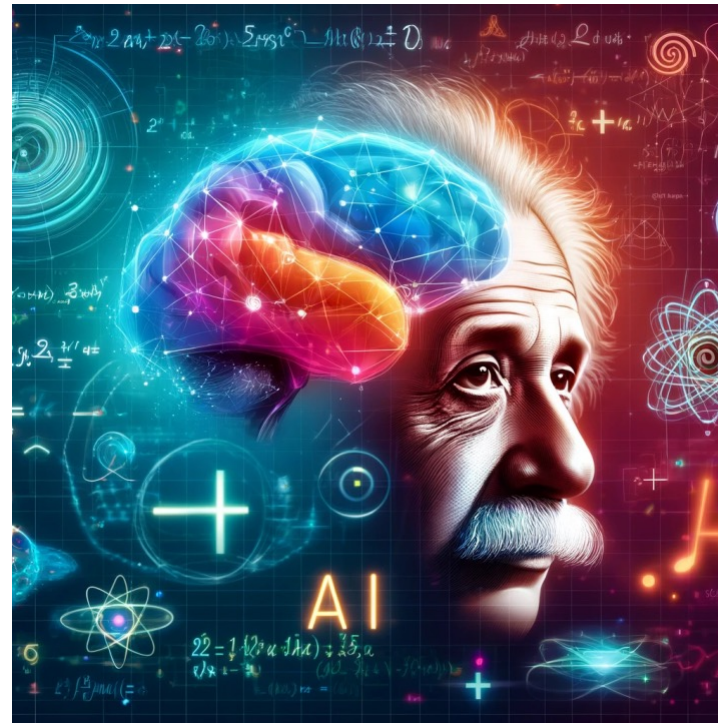


We also have Bellman equation (BE) and Bellman optimality equation (BOE) for the action value $q(s, a)$, and have corresponding TD learning algorithm:

$$q_{t+1}(s_t, a_t) = q_t(s_t, a_t) + \alpha_t(s_t, a_t)[\bar{q}_t - q_t(s_t, a_t)]$$

Algorithm	Equation and expression of $\bar{q}_t$
Sarsa	BE: $q_\pi(s, a) = \mathbb{E}[R_{t+1} + \gamma q_\pi(S_{t+1}, A_{t+1}) S_t = s, A_t = a]$ $\bar{q}_t = r_{t+1} + \gamma q(s_{t+1}, a_{t+1})$
n-step Sarsa	BE: $q_\pi(s, a) = \mathbb{E}[R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^n q_\pi(S_{t+n}, A_{t+n}) S_t = s, A_t = a]$ $\bar{q}_t = r_{t+1} + \gamma r_{t+2} + \cdots + \gamma^n q(s_{t+n}, a_{t+n})$
Q-learning	BOE: $q_\pi(s, a) = \mathbb{E} \left[R_{t+1} + \gamma \max_a q_\pi(S_{t+1}, a) S_t = s, A_t = a \right]$ $\bar{q}_t = r_{t+1} + \gamma \max_a q(s_{t+1}, a)$
Monte Carlo	BE: $q_\pi(s, a) = \mathbb{E}[R_{t+1} + \gamma R_{t+2} + \cdots S_t = s, A_t = a]$ $\bar{q}_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} \dots$

We explore how concepts and insights from science have revolutionized AI



Outline

-  Physics for AI
-  Math for AI
-  Neuroscience for AI
-  Other important insights

I | Physics for AI

1. Principle of least action (最小作用量原理)
2. Path integral (路径积分)
3. Symmetry (对称性)
4. Noether's theorem (诺特定理)
5. Principle of maximum entropy (最大熵原理)
6. Dissipative structure (耗散结构)
7. Emergence (涌现)
8. Phase transitions (相变)

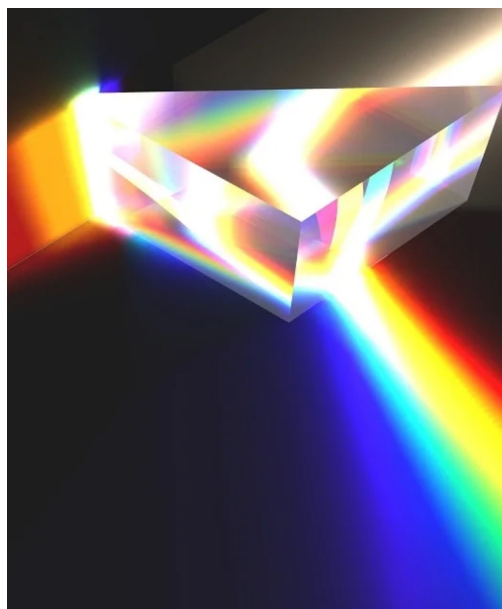


1. Principle of stationary action 最小作用量原理

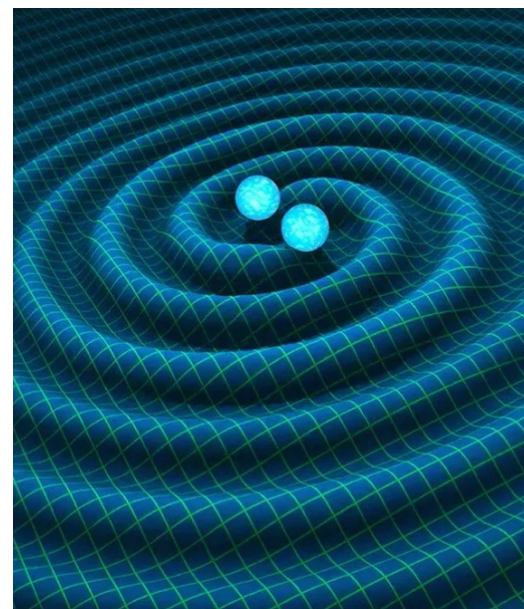
This principle can describe almost all known dynamics, from classical mechanics of how objects move and interact, to how the light reflect and diffract, to Einstein's special and general relativity, and to quantum mechanics.



Classical mechanics



Geometric optics



General relativity



Quantum mechanics

1. Principle of stationary action 最小作用量原理

We all know in classical mechanics, the object's motion is governed by:

$$F = ma$$

Alternatively, we can describe the object's motion using the following simple formula:

$$\delta S = 0$$

which is called **principle of stationary action**.



1. Principle of stationary action 最小作用量原理

Principle of stationary action:

$$\delta S = 0$$

Here S is the action:

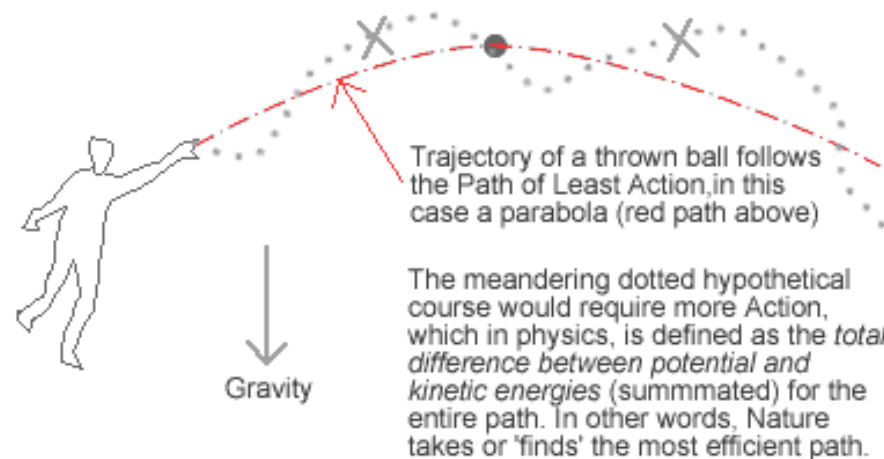
$$S = \int_{t_0}^{t_1} L(q(t), \dot{q}(t), t) dt$$

$q(t)$: position
 $\dot{q}(t)$: velocity
 $\ddot{q}(t)$: acceleration

$L(\cdot)$ is called *Lagrangian*. For classical mechanics,

$$L(q(t), \dot{q}(t), t) = T - V = \frac{1}{2} m \dot{q}(t)^2 - V(q(t))$$

Intuition: For all the possible trajectory $q(t)$ from t_0 to t_1 , Nature choose the one where the action is **stationary**.



1. Principle of stationary action: Euler-Lagrangian equation

Using principle of stationary action $\delta S = 0$ on action $S = \int_{t_0}^{t_1} L(q(t), \dot{q}(t), t) dt$,

we get Euler-Lagrangian equation:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial q}$$

$q(t)$: position
 $\dot{q}(t)$: velocity
 $\ddot{q}(t)$: acceleration

Example: For a spring that is a harmonic oscillator, we have:

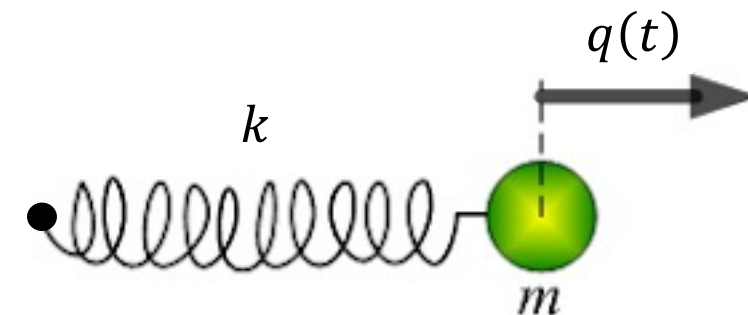
$$L = T - V = \frac{1}{2} m \dot{q}^2 - \frac{1}{2} k q^2$$

Applying the Euler-Lagrangian equation, we have

$$0 = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \frac{d}{dt} (m \dot{q}) - (-kq)$$

$$\text{i.e., } -kq = m\ddot{q}$$

Newton's second law





1. Principle of stationary action: Example

Alternatively, we can obtain Newton's second law directly from $\delta S = 0$.

$$S[q, \dot{q}] = \int_{t_0}^{t_1} \left(\frac{1}{2} m \dot{q}^2 - \frac{1}{2} k q^2 \right) dt$$

$q(t)$: position
 $\dot{q}(t)$: velocity
 $\ddot{q}(t)$: acceleration

Suppose that we have a perturbation $q(t) \rightarrow q(t) + \delta q(t)$ to the trajectory, then

$$\begin{aligned} S[q + \delta q, \dot{q} + \delta \dot{q}] &= \int_{t_0}^{t_1} \left(\frac{1}{2} m (\dot{q} + \delta \dot{q})^2 - \frac{1}{2} k (q + \delta q)^2 \right) dt \\ &= S[q, \dot{q}] + \int_{t_0}^{t_1} m \dot{q} \delta \dot{q} dt - \int_{t_0}^{t_1} k q \delta q dt \\ &= S[q, \dot{q}] + \left(m \dot{q} \delta q \Big|_{t_0}^{t_1} - \int_{t_0}^{t_1} m \ddot{q} \delta q dt \right) - \int_{t_0}^{t_1} k q \delta q dt \\ &= S[q, \dot{q}] + \int_{t_0}^{t_1} (-m \ddot{q} - k q) \delta q dt \end{aligned}$$

The principle stipulate that $\delta S = \int_{t_0}^{t_1} (-m \ddot{q} - k q) \delta q dt = 0$, regardless of δq . Then we have

$$-kq = m\ddot{q}$$

Newton's second law



1. Principle of stationary action: Fermat's principle

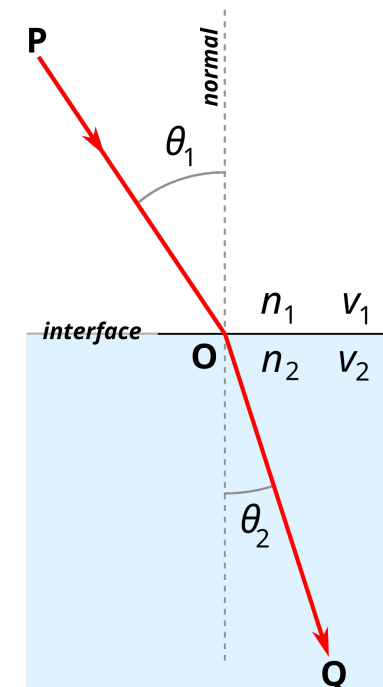
Why does the light refraction (折射) when going through a second median with a different refraction index?

We know the Snell's law: $\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_1}{n_2}$

There is also a more general principle for geometric optics:

Fermat's principle: The path taken by a ray between two given points is the path that can be traveled in the least time.

This is a direct consequence of the **principle of least action**.



General relativity:

$$S[g] = \frac{1}{2\kappa} \int R \sqrt{-g} d^4x$$

Quantum mechanics: Feynman's path integral (will introduce in a moment)



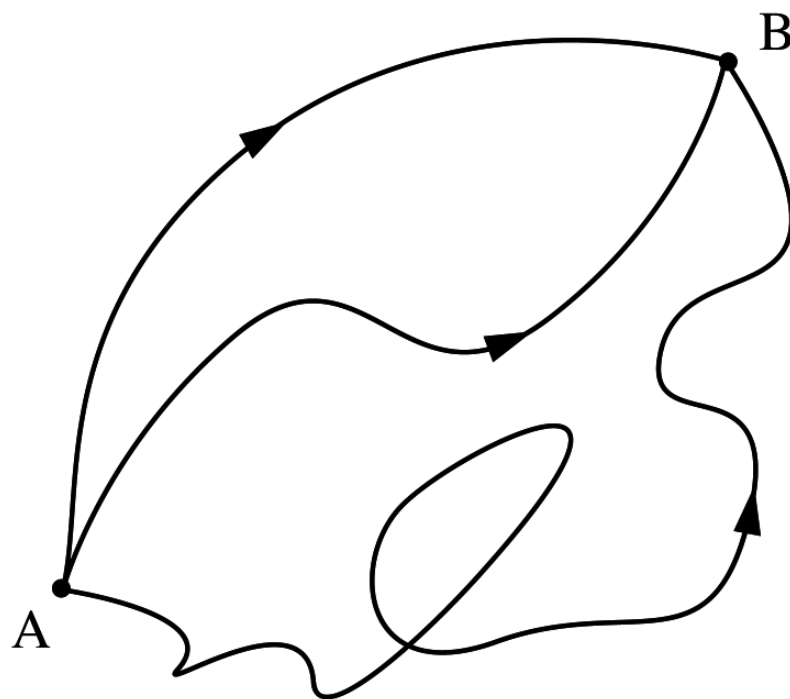


1. Principle of stationary action: Insight for AI

- This variational principle is a very powerful technique that can unify lots of seemingly different phenomenon.
- **Some applications**
 - Cranmer, Miles, et al. "Lagrangian neural networks." *arXiv preprint arXiv:2003.04630* (2020).
 - Finzi, Marc, et al. "Simplifying hamiltonian and lagrangian neural networks via explicit constraints." NeurIPS 2020



Path integral essentially states that in quantum mechanics, to go from A to B, a particle will explore all possible paths.



To go from A to B, path integral states that it will explore all possible paths



Recall that in quantum mechanics, a quantum state is represented by a wavefunction. For example, $|\psi\rangle = \frac{1}{\sqrt{2}}|000000\rangle + \frac{1}{\sqrt{2}}|001001\rangle$

A **propagator** of a quantum system between two spacetime points (x', t') and (x_0, t_0) is the probability transition amplitude between the wavefunction evaluated at those points:

$$U(x', t'; x_0, t_0) = \langle \psi(x', t') | \psi(x_0, t_0) \rangle$$

Path integral states:

$$U(x', t'; x_0, t_0) = \frac{1}{Z} \int_{x(0)=x_0, x(t')=x'} \mathcal{D}\mathbf{x} e^{iS[\mathbf{x}, \dot{\mathbf{x}}]}$$

where $\mathcal{D}\mathbf{x}$ denotes integration over all paths, $S[\mathbf{x}, \dot{\mathbf{x}}] = \int L(\mathbf{x}, \dot{\mathbf{x}}) dt$.

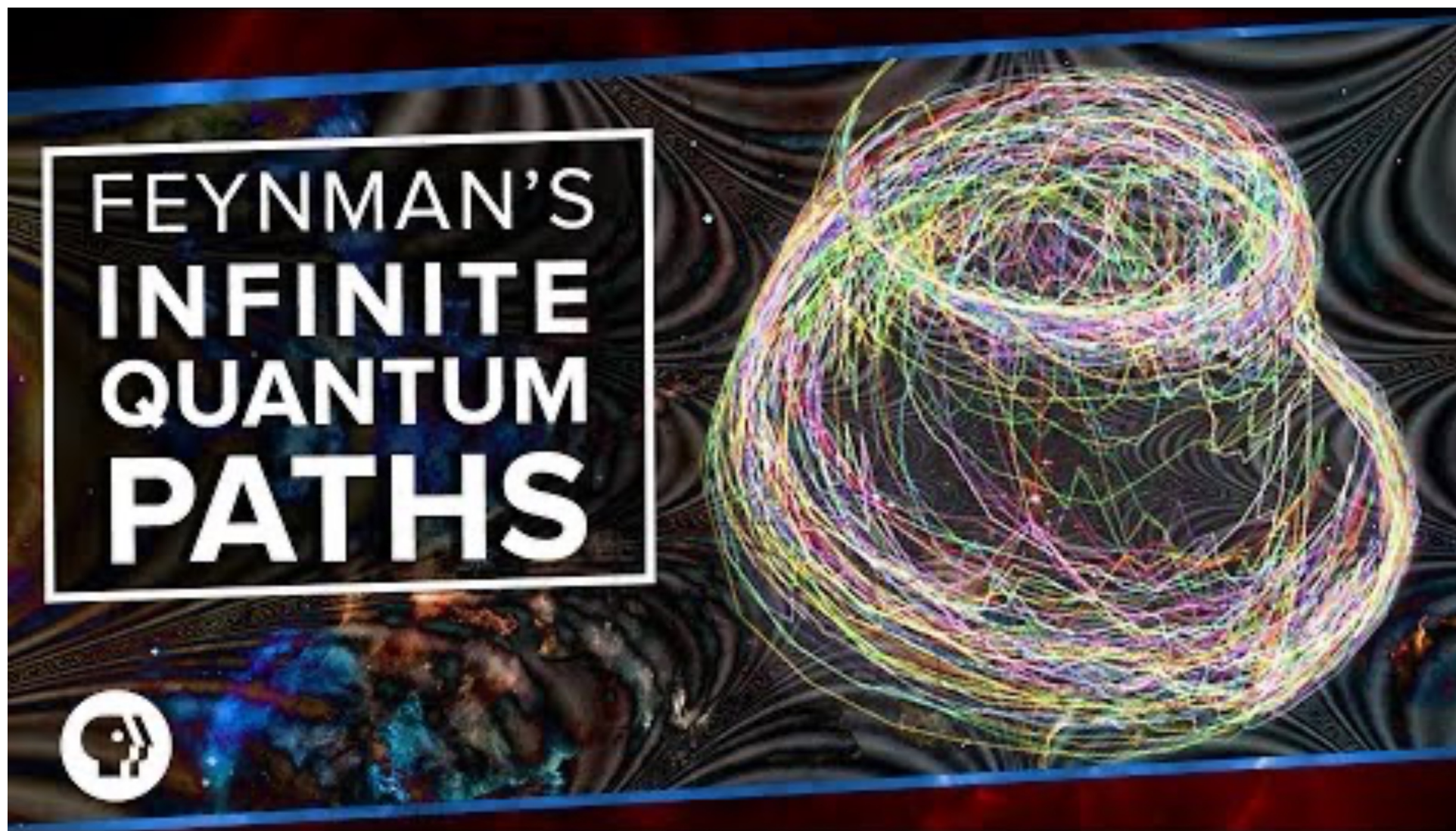




2. Path integral



WESTLAKE
UNIVERSITY



<https://www.youtube.com/watch?v=vSFRN-ymfgE>



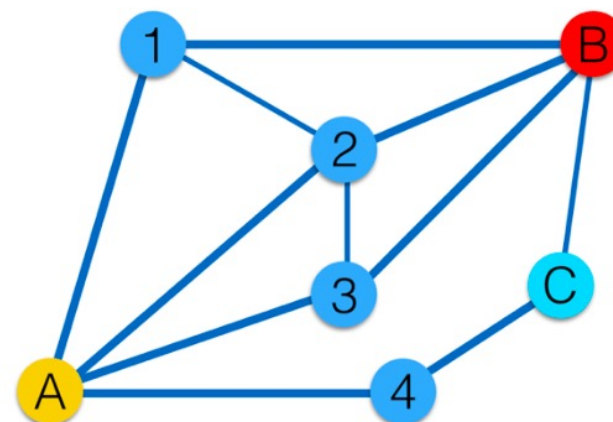
2. Path integral: Application in AI

- Path Integral Based Convolution and Pooling for Graph Neural Networks, NeurIPS 2020



$$\int \rightarrow \Sigma$$

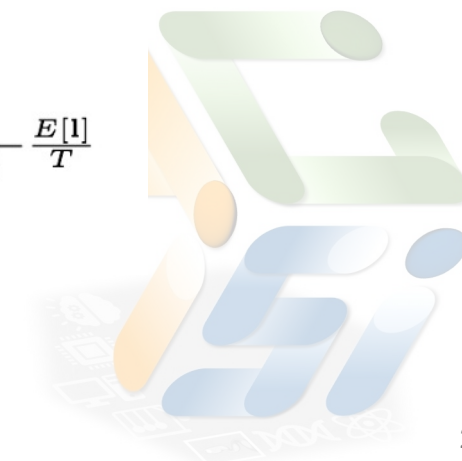
$$e^{iS[\mathbf{x}, \dot{\mathbf{x}}]} \rightarrow e^{-\frac{E[\mathbf{l}]}{T}}$$



$$\phi(x, t) = \frac{1}{Z} \int \phi_0(\mathbf{x}) \int \mathcal{D}(\mathbf{x}) e^{iS[\mathbf{x}, \dot{\mathbf{x}}]}$$

$$\phi_i = \frac{1}{Z_i} \sum_{j=1}^N \phi_j \sum_{\{\mathbf{l} | l_0=i, l_{|1|}=j\}} e^{-\frac{E[\mathbf{l}]}{T}}$$

Influence



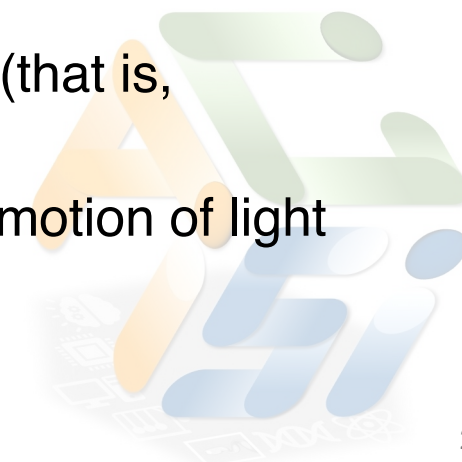
Physics is essentially the study of **symmetry**: Features of the system (observed or intrinsic) that is preserved or remains unchanged under some transformation

Originally, physicists perform research through **induction**: Summarizing laws from data.

Einstein pioneers a new paradigm by **first defining the symmetry** the system obeys (as axioms), and then derive the theory for the system.

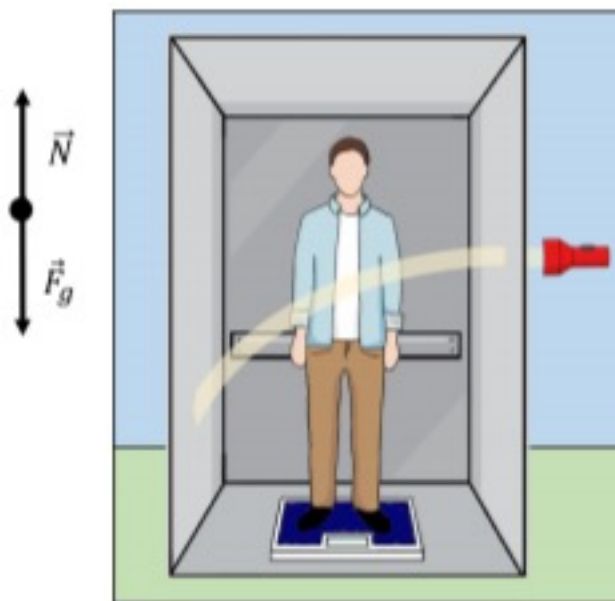
Special relativity starts from the following two symmetries:

- The laws of physics are invariant (identical) in all inertial frames of reference (that is, frames of reference with no acceleration).
- The speed of light in vacuum is the same for all observers, regardless of the motion of light source or observer.

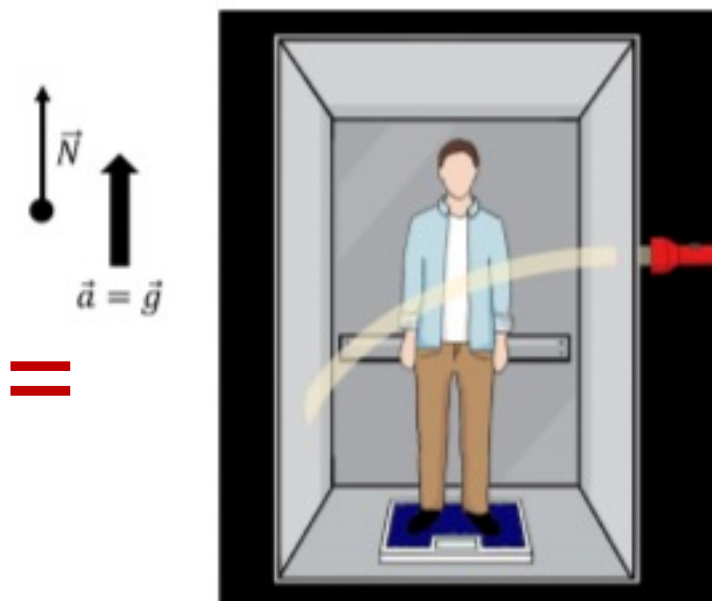


General relativity starts from the following symmetry:

- The outcome of any local, non-gravitational test experiment is independent of the experimental apparatus' velocity relative to the gravitational field and is independent of where and when in the gravitational field the experiment is performed.



In gravitational field



In an accelerating elevator



Almost **all neural architecture** can be defined starting from **symmetry**.

Equivariance (等变) : If $f(g x) = g \circ f(x)$ where g is a transformation, then the function f is equivariant under g .

Invariance (不变) : If $f(g x) = f(x)$, then the function f is invariant under g .

- Translational equivariance $\Rightarrow$ Convolutional Neural Network (CNN)
- Permutation equivariance $\Rightarrow$ Graph Neural Network (GNN)
- Permutation equivariance + translation/rotational equivariance $\Rightarrow$ Equivariant GNN [1]
- Translational equivariant position embedding $\Rightarrow$ Rotary Position Embedding [2]

[1] Satorras, Victor Garcia, et al. "E (n) equivariant graph neural networks." ICML 2021

[2] Su, Jianlin, et al. "Roformer: Enhanced transformer with rotary position embedding." *Neurocomputing* 568 (2024): 127063.

Noether's theorem states that every continuous symmetry of the action of a physical system with conservative forces has a corresponding conservation law.

- **Rotation symmetry $\Leftrightarrow$ conservation of angular momentum:** If a physical system behaves the same regardless of how it is oriented in space (that is, it's invariant), its Lagrangian is symmetric under continuous rotation: from this symmetry, Noether's theorem dictates that the angular momentum of the system be conserved
- **Space translation symmetry $\Leftrightarrow$ linear momentum conservation**
- **Time translation symmetry $\Leftrightarrow$ energy conservation**



Noether's Learning Dynamics: Role of Symmetry Breaking in Neural Networks, NeurIPS 2021

- It models the discrete learning dynamics of gradient descent using a continuous-time Lagrangian formulation, in which the learning rule corresponds to the kinetic energy and the loss function corresponds to the potential energy.
- Then, the authors identify kinetic symmetry breaking (KSB), the condition when the kinetic energy explicitly breaks the symmetry of the potential function. We generalize Noether's theorem known in physics to take into account KSB and derive the resulting motion of the Noether charge: Noether's Learning Dynamics (NLD)



It states that the probability distribution which best represents the current state of knowledge about a system is the one with **largest entropy**, in the context of precisely stated prior data.

The entropy of a probability distribution is defined as

$$H(p) = - \sum_{i=1}^n p_i \log p_i$$

For example, if you have a dice, and don't know anything about its property, then the most natural assumption is that each face will land with 1/6 probability, resulting in a maximum entropy of $(\log_2 6)$ bits.



Other examples:

- **Gaussian distribution:** $\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

This is the distribution that maximizes the entropy for
 $\mathbb{E}[x] = \mu, \text{Var}(x) = \sigma^2$

- **Exponential distribution:** $p(x; \lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$

This is the distribution that maximizes the entropy if
 $\mathbb{E}[x] = \frac{1}{\lambda}$ and it is supported on $[0, +\infty)$



Application in Reinforcement Learning (RL):

- Using entropy regularization to encourage exploration

- Standard RL objective: Maximize

$$\mathbb{E}_{\pi}[r_t]$$

- Soft actor-critic (SAC) objective: Maximize

$$\mathbb{E}_{\pi}[r_t + H(\pi(a|s))]$$



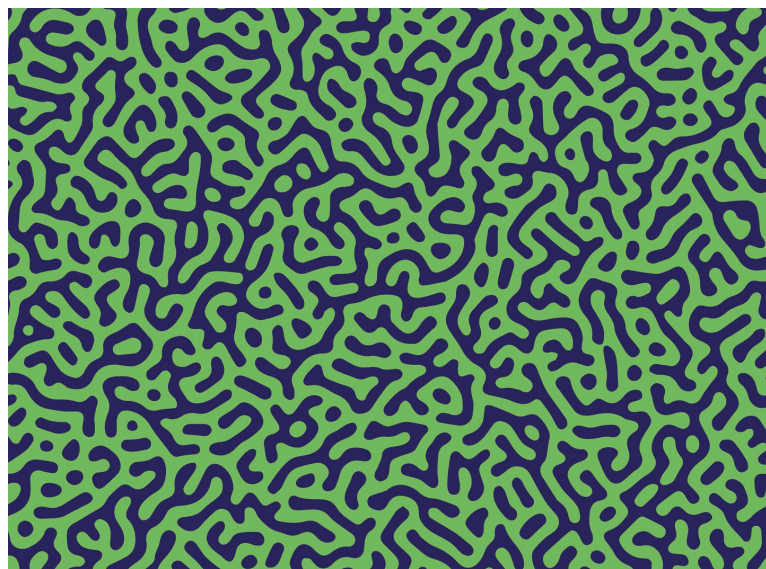
A 200 ml Pyrex beaker containing a red liquid. A red horizontal bar is positioned at the top of the beaker. The beaker has a white label with a red circle and the text "200 ml PYREX®". The liquid level is marked with a red line at the 100 ml mark. The beaker is sitting on a white surface. In the bottom right corner, there is a black box with the text "MakeAGIF.com".



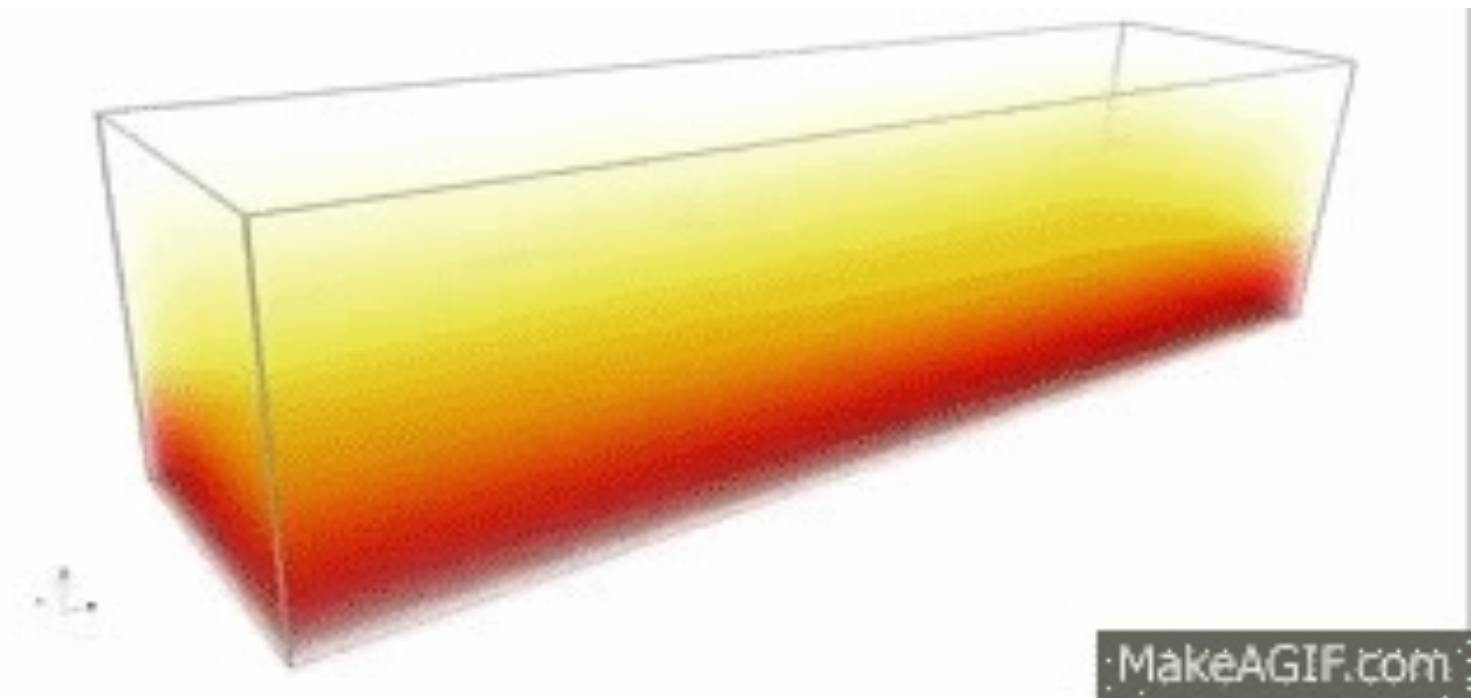
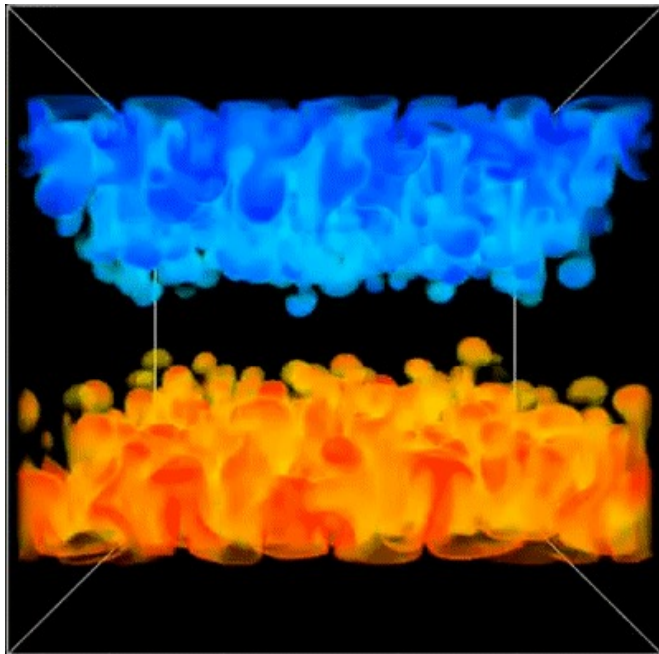


6. Dissipative structures: Turing pattern

Turing pattern (图灵斑图) :



Rayleigh–Bénard convection (贝纳尔对流) :



MakeAGIF.com





6. Dissipative structures: Key features

- **Occurs in open systems** that require continuous input of energy or matter from the external environment to sustain themselves. More fundamentally, they depend on a "negative entropy flux."
- **Emerges only when control parameters reach critical thresholds**, meaning dissipative structures arise exclusively far from thermodynamic equilibrium. This "far-from-equilibrium" condition must extend beyond the linear regime governed by irreversible processes into nonlinear territory.
- **Exhibits spatiotemporal patterns** with lower symmetry than the initial homogeneous state. In other words, dissipative structures emerge through spontaneous symmetry breaking.
- **Demonstrates remarkable stability**, remaining robust against small perturbations.





6. Dissipative structures: Significance

- Dissipative structures serve as a conceptual bridge between **physical sciences** and **life sciences**.
- While life's emergence and species evolution—progressing from simplicity to complexity, disorder to order—appear to contradict the Second Law of Thermodynamics, dissipative structure theory resolves this paradox.
- Its originator, Ilya Prigogine, was awarded the 1977 Nobel Prize in Chemistry for this groundbreaking work.



Emergence occurs when a complex entity has properties or behaviors that its parts do not have on their own, and emerge only when they interact in a wider whole.

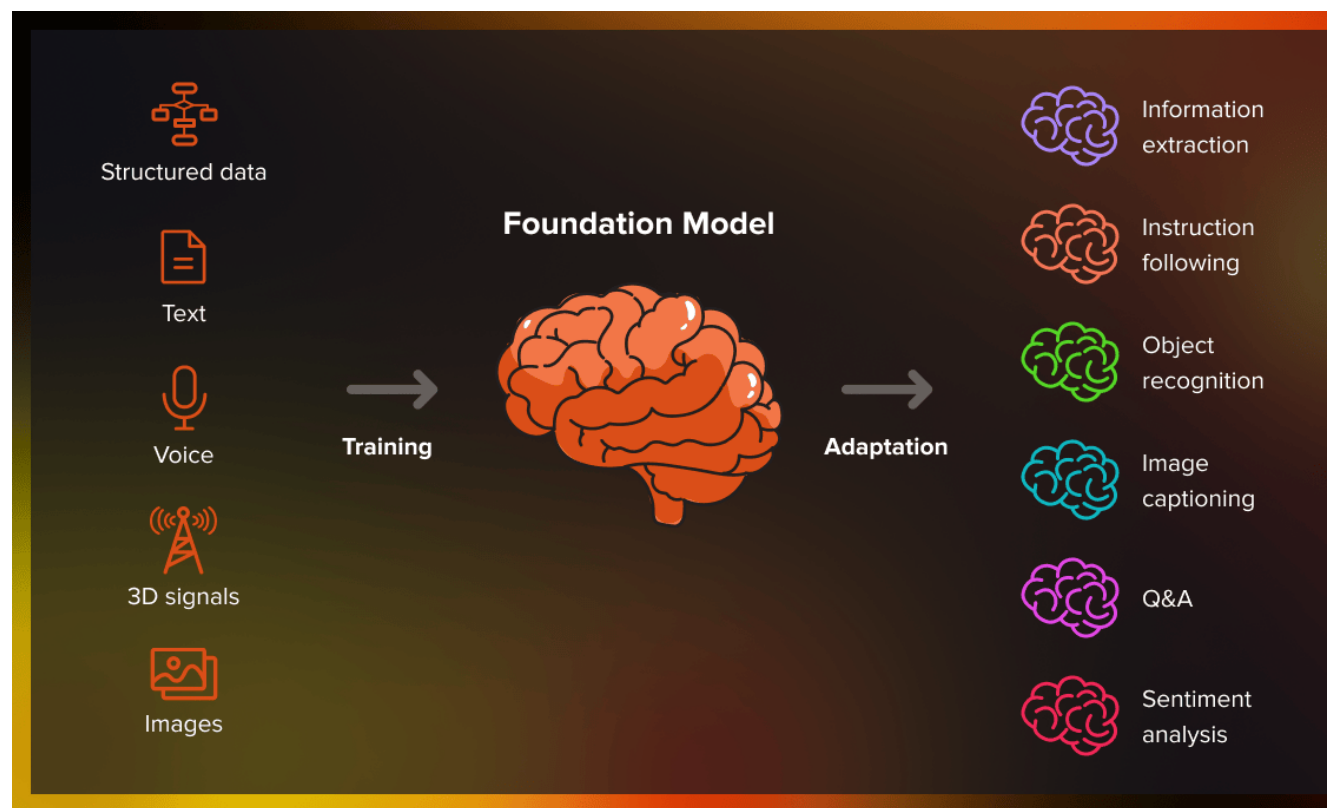


flocks (鸟群)



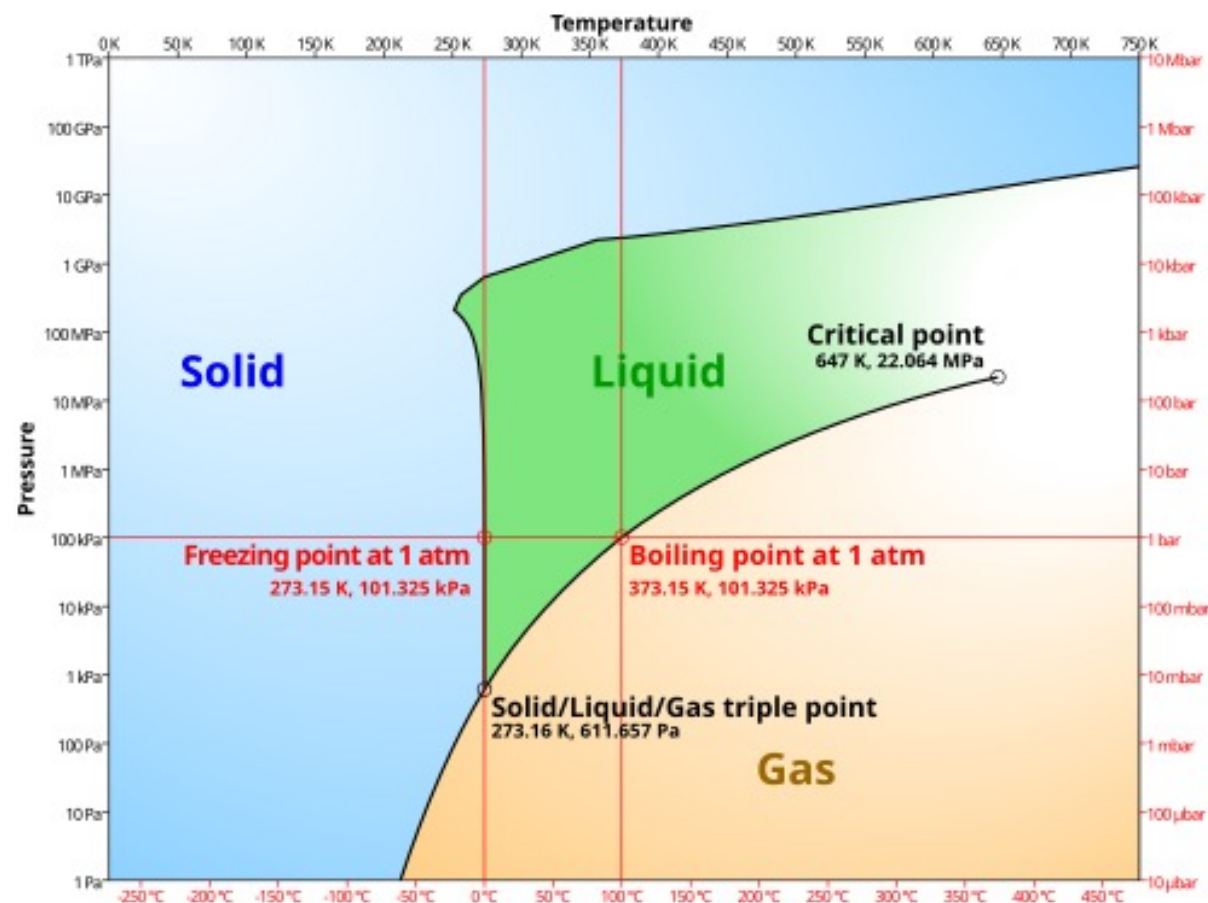
brain

Foundation models also exhibit **emergence** when trained with large amount of data and with enough model size and training iterations



8. Phase transitions (相变)

Phase transition is the physical process of transition between one state of a medium and another.





8. Phase transitions: Key characteristics

- **Order parameters**
 - A measure of the degree of order across the boundaries in a phase transition system; it normally ranges between zero in one phase (usually above the critical point) and nonzero in the other.
- **Symmetry breaking**
 - Phase transitions often involve a symmetry breaking process
- **Critical exponents and universality classes**
$$C \propto |T - T_c|^{-\alpha}$$
- **Critical behavior**
 - May observe critical slowing down or speeding up



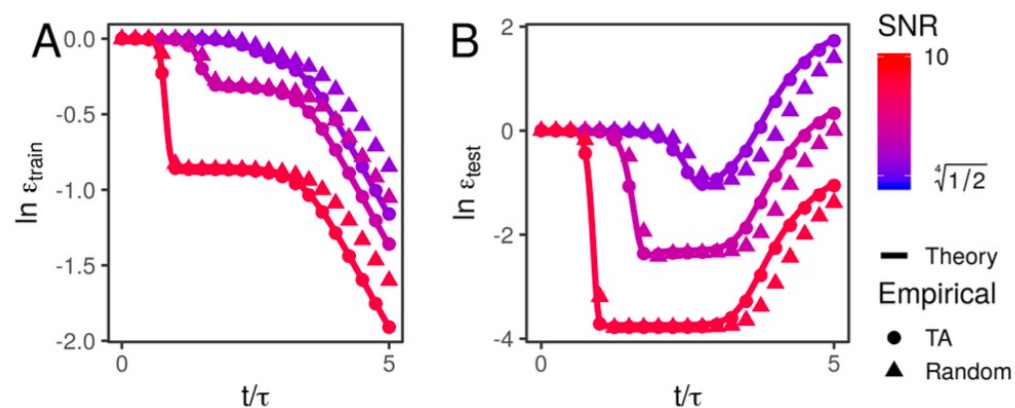


8. Phase transitions in learning

- An analytic theory of generalization dynamics and transfer learning in deep linear networks, ICLR 2019

Provides analytic solutions to the training and testing error of deep networks as a function of training time, number of examples, network size and initialization for deep linear networks

Reveals that deep networks progressively learn the most important task structure first, so that generalization error at the early stopping time primarily depends on task structure and is independent of network size.



II

Math for AI

1. Game theory (博弈论)
2. Integral transformations (积分变换)



1. Game theory: Guess 1/3 of average number

Each player: Choose an integer between 0 and 100. The player closest to **1/3 of the average** of the numbers wins the prize.

First time play: 99+15+11+7+1+15+9+20. Winner: 7

Second time play: 1+3+5+13+11+5+3+2. Winner: 2

$$(3+13+7+2+23+63+66+43+1+33+73+10)/12/3 = 10$$

$$(6+6+3+5+4+2+33+4+3+2+5+5)/12/3 = 2$$



1. Game theory (博弈论)



Static/Dynamic (Simultaneous/Sequential):

- *Simultaneous-move*: Players act concurrently without observing others' choices
- *Sequential-move*: Players act after observing preceding actions

Perfect/Imperfect Information:

- *Perfect information*: Players have complete knowledge of all prior events when making decisions, including initial conditions (e.g., starting hands in card games). Examples: Chess.
- *Imperfect information*: Players lack full knowledge of previous moves or states

Zero-sum/Non-zero-sum:

- *Zero-sum*: Players' payoffs sum to zero, creating purely competitive scenarios with no cooperation incentives
- *Non-zero-sum*: Payoff sums vary across strategy profiles, enabling potential cooperation



Symmetric/Asymmetric:

- *Symmetric*: Players share identical objectives and payoff structures (outcomes depend solely on strategy choices). Example: Chess (identical win conditions for both players)
- *Asymmetric*: Players have divergent objectives and evaluation metrics
 - Example: Business negotiations between buyers/sellers

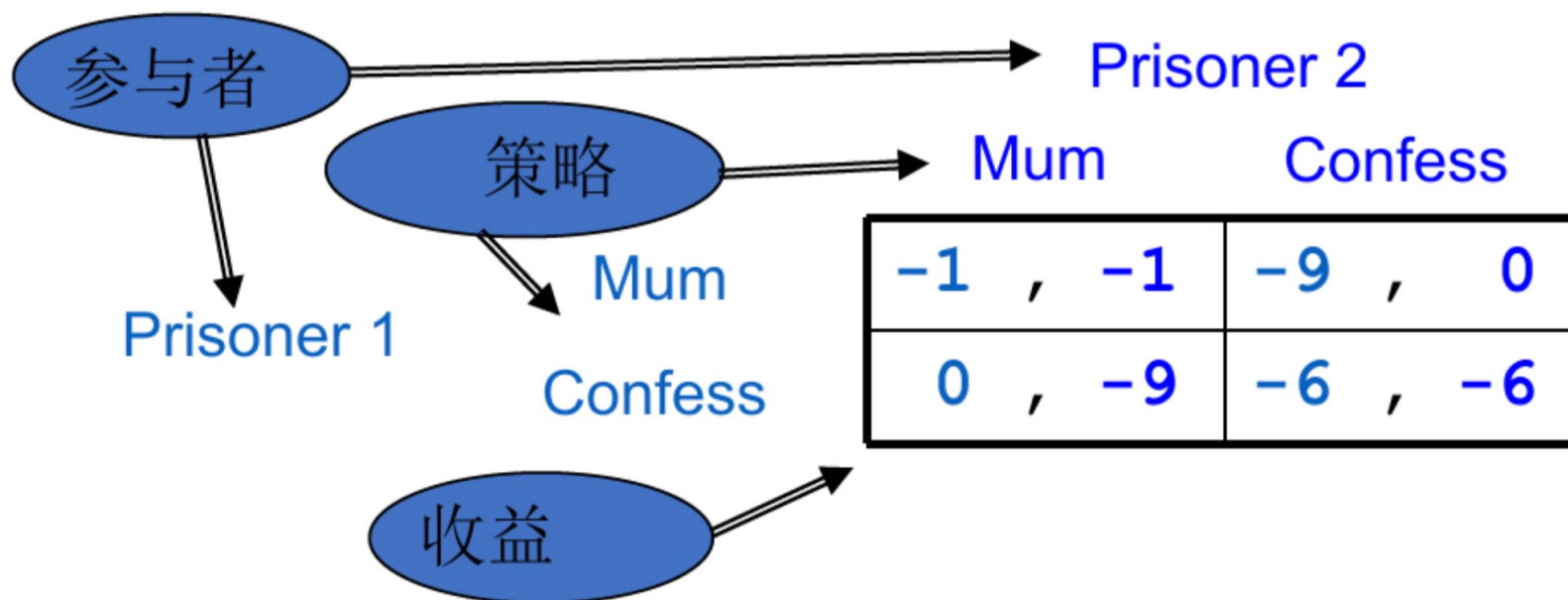
Cooperative/Non-cooperative:

- *Cooperative*: Players form binding alliances through enforceable agreements
- *Non-cooperative*: Players cannot form binding alliances - any cooperation must emerge through self-enforcing mechanisms



1. Game theory: Prisoner's dilemma

- Set of Participants: {Prisoner 1, Prisoner 2}
- Decision set $S_1 = S_2 = \{\underline{\text{Mum}}, \underline{\text{Confess}}\}$
- Utility function u_1, u_2 :





Rationality Consensus (理性共识)

- Zeroth-order rationality (零阶理性) : Each agent maximizes their own payoff
- First-order rationality (一阶理性) : 零阶理性基础上, 每个人知道其他参与者是理性的
- Second-order rationality (二阶理性) : 一阶理性基础上, 每个人知道其他人知道自己是理性的
-
- nth-order rationality (n阶理性)



Nash equilibrium: In a standard game $\{S_1, S_2, \dots, S_n, u_1, u_2, \dots, u_n\}$, a policy combination $(s_1^*, s_2^*, \dots, s_n^*)$ is **Nash equilibrium**, if for each participant i :

$$u_i(s_1^*, s_2^*, \dots, s_{i-1}^*, s_i^*, s_{i+1}^*, \dots, s_n^*) \geq u_i(s_1^*, s_2^*, \dots, s_{i-1}^*, s_i, s_{i+1}^*, \dots, s_n^*)$$

for all $s_i \in S_i$.

Nash equilibrium is a **stationary** result, but not necessarily the best result.





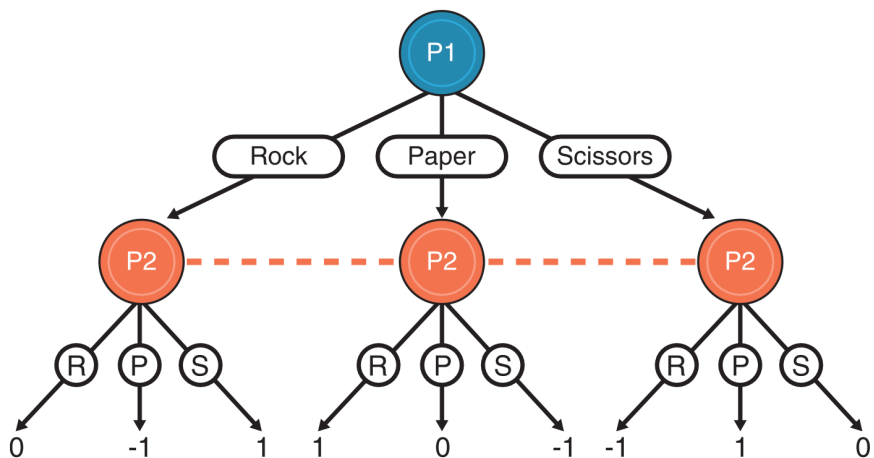
1. Game theory: Prisoner's dilemma revisited

- Set of Participants: {Prisoner 1, Prisoner 2}
- Decision set $S_1 = S_2 = \{\underline{\text{Mum}}, \underline{\text{Confess}}\}$
- Utility function u_1, u_2 :

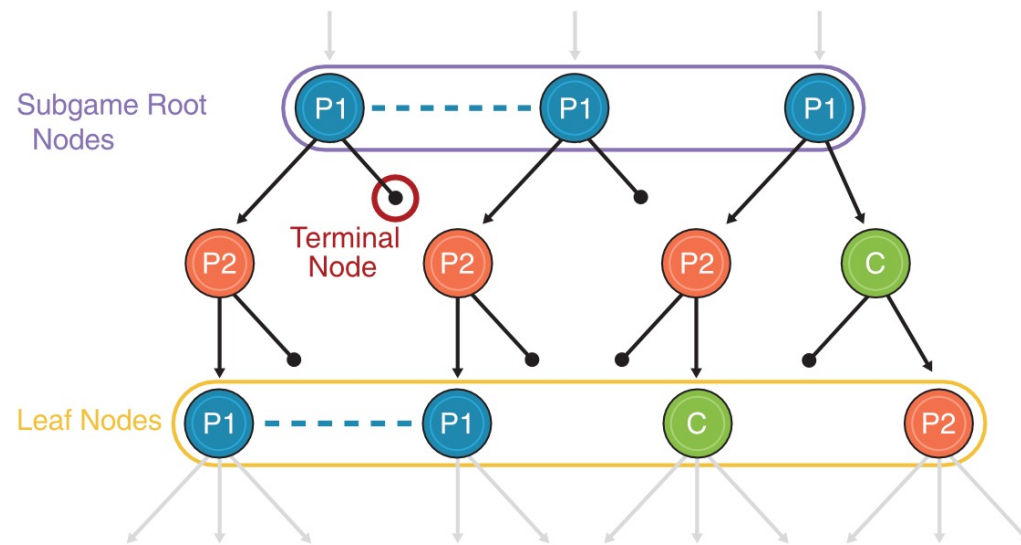
		Prisoner 2	
		Mum	Confess
Prisoner 1	Mum	-1 , -1	-9 , <u>0</u>
	Confess	<u>0</u> , -9	<u>-6</u> , <u>-6</u>



- Superhuman AI for multiplayer poker, Science 2019



Rock-paper-scissors



Poker

Integral transforms:

$$F(k) = \int_a^b K(k, x) f(x) dx$$

- Fourier transform: $K(k, x) = e^{-ikx}$, $-\infty < x < \infty$
- Laplace transform: $K(k, x) = e^{-kx}$, $0 \leq x < \infty$
- Sine transform: $K(k, x) = \sin(kx)$, $-\infty \leq x < \infty$
- Cosine transform: $K(k, x) = \cos(kx)$, $-\infty \leq x < \infty$
- Hankel transform: $K(k, x) = xJ_n(kx)$, $0 \leq x < \infty$
- Merlin transform: $K(k, x) = x^{k-1}$, $0 \leq x < \infty$



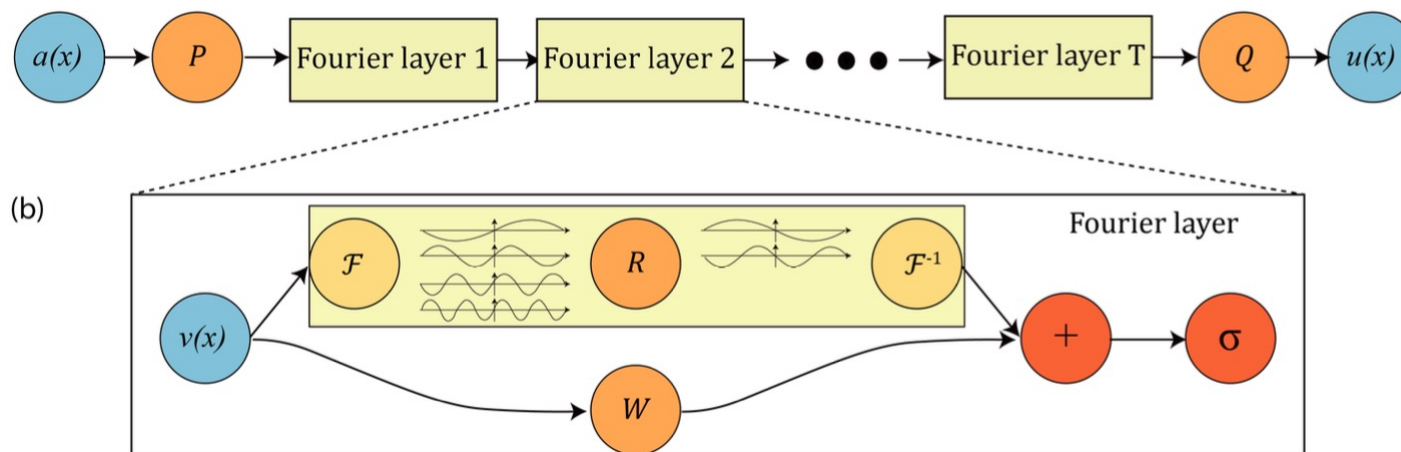


2. Integral transforms: Useful properties 1

Integral transforms can **transform convolution to product in the frequency space**:

$$\int_D \kappa(x - y) v_t(y) dy = \mathcal{F}^{-1}(\mathcal{F}(\kappa) \cdot \mathcal{F} v_t)(x)$$

This is the foudation for Fourier Neural Operator (FNO) [1]



$$v_{t+1}(x) := \sigma \left(\underbrace{W v_t(x)}_{\text{非线性}} + \underbrace{(\mathcal{K}(a; \phi) v_t)(x)}_{\text{全局积分}} \right)$$

非线性 局域变换 全局积分

$$(\mathcal{K}(a; \phi) v_t)(x) := \mathcal{F}^{-1}(\mathcal{R}_\phi \cdot \mathcal{F} v_t)(x)$$





2. Integral transforms: Useful properties 2

Integral transforms can transform a differential equation to an algebra equation

Suppose that we have an Ordinary Differential Equation (ODE)

$$\frac{du(t)}{dt} = f(u(t))$$

Transforming both sides to the Laplace space, we have:

$$pU(p) - u(0) = F(U(p))$$

where $F(U(p))$ is the Laplace transform of $f(u(t))$.



II

Neuroscience for AI



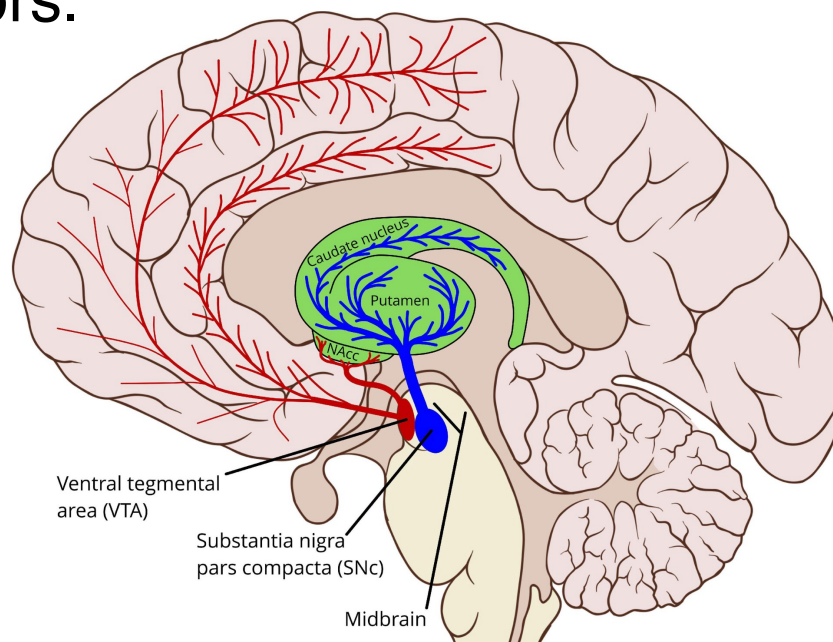
1. Neural networks
2. Convolutional Neural Networks
 1. Inspired by the hierarchical processing in the visual cortex
3. Attention
 1. Inspired by the brain's ability to focus on salient stimuli
4. **Reinforcement learning**
5. **Synaptic plasticity**
6. **Memory-Augmented Networks**
7. **Grid cells**
8. **Edge of chaos**



Temporal difference learning for the action value:

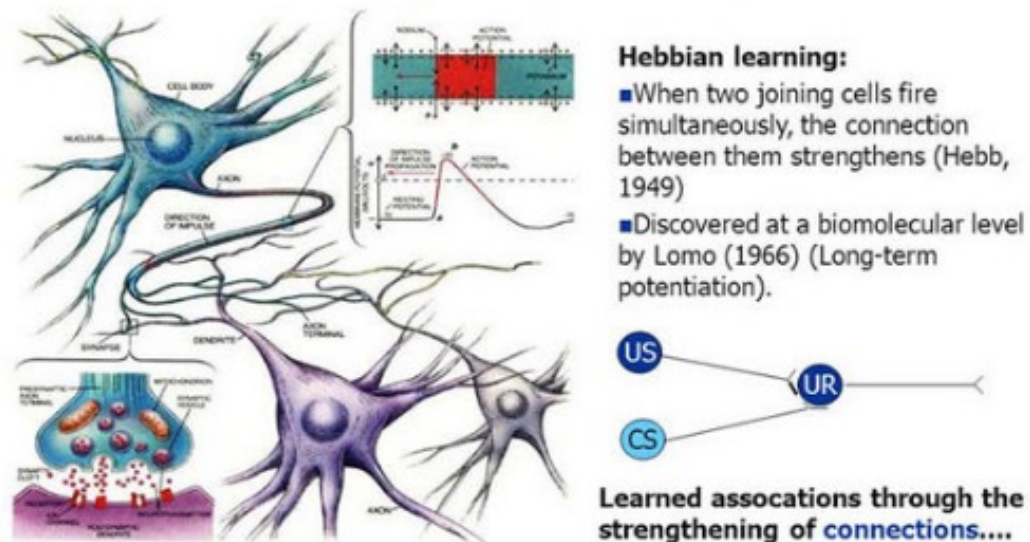
$$q_{t+1}(s_t, a_t) = q_t(s_t, a_t) + \alpha_t(s_t, a_t)[\bar{r}_t - q_t(s_t, a_t)]$$

It is inspired by the brain's dopaminergic (多巴胺) system, which encodes reward prediction errors.



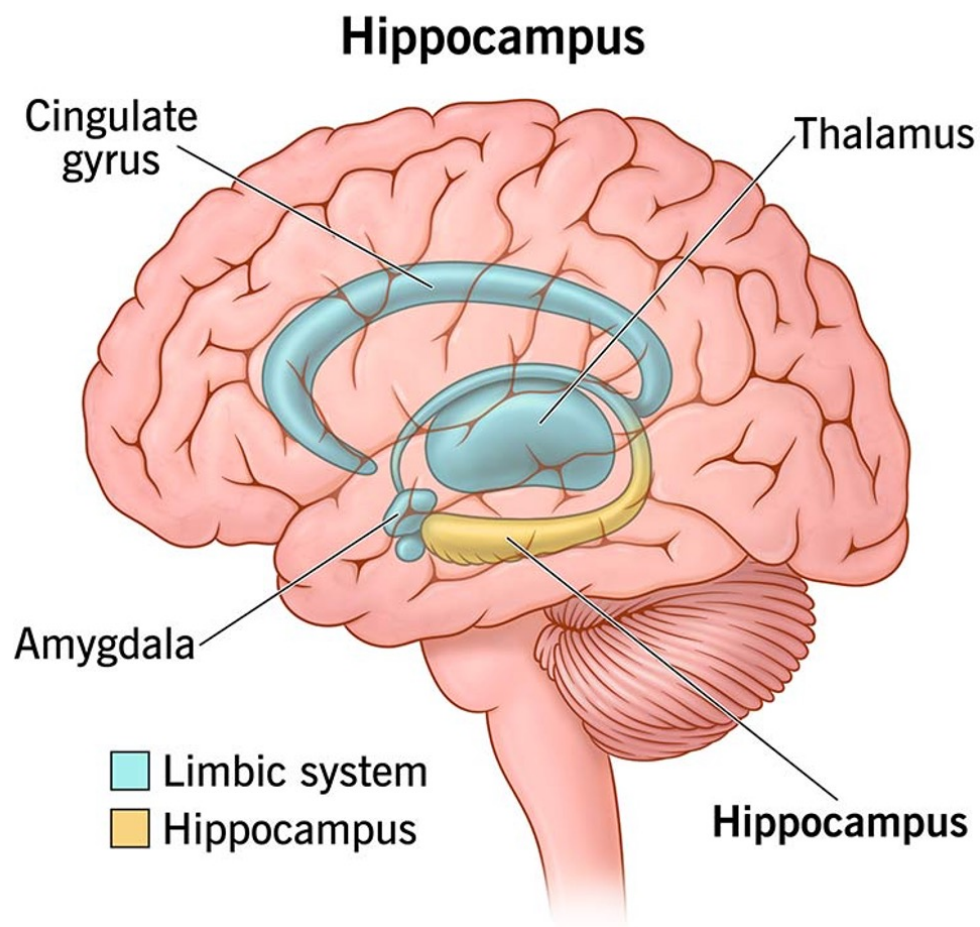
The brain's ability to strengthen/weaken connections (e.g., Hebbian learning: "cells that fire together, wire together") influenced weight adjustment in neural networks.

Hebbian Learning



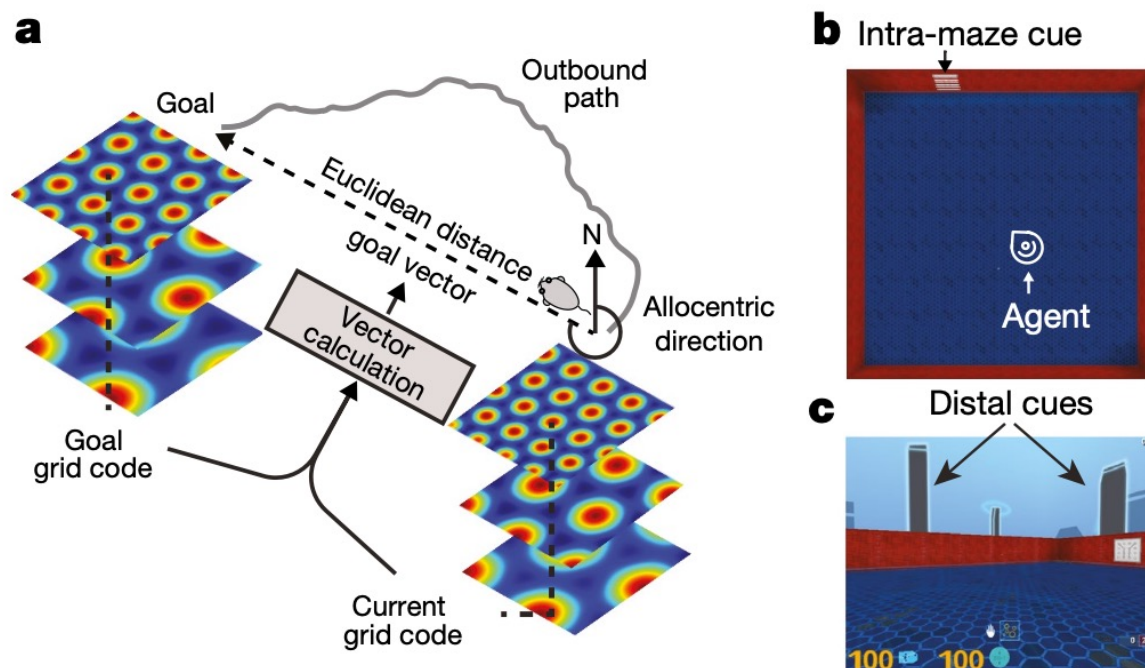
6. Memory-augmented networks

The hippocampus's (海马体) role in episodic memory inspired architectures like LSTMs and Neural Turing Machines



7. Grid cells

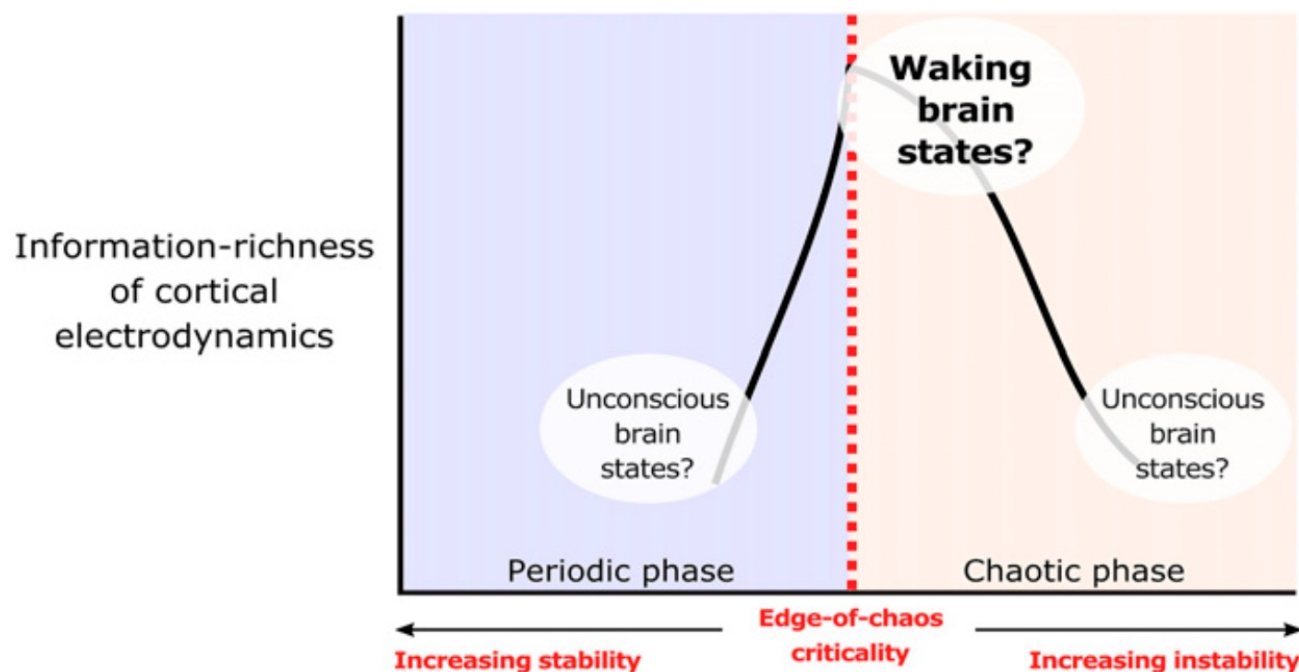
- Mammals use grid cells in the entorhinal cortex to perform navigation.
- It provides a multi-scale periodic representation that functions as a metric for coding space.
- Using deep RL, DeepMind researchers [1] demonstrate grid-cell like representation developed in neural networks



[1] Banino, Andrea, et al. "Vector-based navigation using grid-like representations in artificial agents." *Nature* 557.7705 (2018): 429-433.

8. Edge of chaos

- The dynamics of the cortex may be poised near the edge-of-chaos critical point.
- This near-critical behavior supports the vast flow of information through cortical networks during conscious states.



IV | Other important insights

Concepts	Source	Application
Diffusion	physics	Diffusion models
Energy	physics	Energy-based models, Boltzmann machines
Hamiltonian	physics	Hamiltonian Monte Carlo, Hamiltonian Neural Networks
Working memory	neuroscience	Working Memory Graphs
Graph theory	math	Graph Neural Networks
Deep Ridgelet transform	math	
Control theory	Engineering	
Free energy principle	Neuroscience	
Renormalization	physics	

- 100人生: 物理学解锁世界的100个顶级思维模型

